

Problem 7

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Problem 1 Let $f(x) = \frac{1}{1+x^2}$. Find the following for f :

- (a) f' and f''

Explanation.

$$\begin{aligned} f'(x) &= \frac{(1+x^2)(0) - 1(2x)}{(1+x^2)^2} \\ &= \frac{-2x}{(1+x^2)^2} \end{aligned}$$

$$\begin{aligned} f''(x) &= \frac{(1+x^2)^2(-2) - (-2x)(2)(1+x^2)(2x)}{(1+x^2)^4} \\ &= \frac{-2(1+x^2) + 8x^2}{(1+x^2)^3} \\ &= \frac{6x^2 - 2}{(1+x^2)^3} \end{aligned}$$

- (b) *Critical points*

Explanation. Since $1+x^2 > 0$ for all x , f is differentiable over all real numbers. Thus all critical points of f occur when $f'(x) = 0$. But a fraction equals 0 if and only if its numerator equals 0. So

$$f'(x) = 0 \quad \implies \quad -2x = 0 \quad \implies \quad x = 0$$

Hence, the only critical point is $x = 0$.

- (c) *Local extrema (and check your answers with both the first and second derivative tests)*

Explanation. At $x = 0$, f' changes sign from positive to negative. Thus f goes from increasing to decreasing, and therefore by the first derivative test $x = 0$ is a local maximum of f .

For the second derivative test, we have that

$$f''(0) = \frac{6(0)^2 - 2}{(1+0^2)^3} = \frac{-2}{1} = -2 < 0$$

and thus we again conclude that $x = 0$ is a local maximum of f .

(d) *Inflection points. EXPLAIN.*

Explanation. By the results in part

$$\begin{aligned} f''(x) &= \frac{6x^2 - 2}{(1 + x^2)^3} \\ &= \frac{6(x^2 - \frac{1}{3})}{(1 + x^2)^3} \\ &= \frac{6(x - \frac{1}{\sqrt{3}})(x + \frac{1}{\sqrt{3}})}{(1 + x^2)^3} \end{aligned}$$

which implies that the only candidates for inflection points are points where $x = \pm \frac{1}{\sqrt{3}}$. On the other hand, since $-1 < -\frac{1}{\sqrt{3}} < 0$, $f''(-1) = \frac{1}{2}$, and $f''(0) = -2$, it follows that f'' changes the sign at $x = -\frac{1}{\sqrt{3}}$. Similarly, since $0 < x = \frac{1}{\sqrt{3}} < 1$, $f''(0) = -2$, $f''(1) = \frac{1}{2}$, it follows that f'' changes the sign at $x = \frac{1}{\sqrt{3}}$, too. Since,

$$\begin{aligned} f\left(\frac{1}{\sqrt{3}}\right) &= \frac{3}{4} \\ f\left(-\frac{1}{\sqrt{3}}\right) &= \frac{3}{4} \end{aligned}$$

The function f has inflection points at $\left(-\frac{1}{\sqrt{3}}, \frac{3}{4}\right)$, $\left(\frac{1}{\sqrt{3}}, \frac{3}{4}\right)$ because f is continuous at those points and f changes concavity around those points.